MODEL QUESTION PAPER

REVISION-2010
SUB CODE: 3058

Reg. No…………………
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THIRD SEMESTER DIPLOMA IN ENGINEERING/TECHNOLOGY

ELECTRONICS CIRCUITS
(Common to AE, BM, EA, EL, EC, EP, MD, TC)

(Maximum Marks-100) Time: 3 Hrs.

PART-A

(Answer all questions in one or two sentences. Each question carries 2 Marks)

I

1. Define bandwidth of an amplifier.
2. Write the expression for resonant frequency of series resonant circuit.
3. Mention Barkhausen Criteria for oscillation'
4. Draw an RC integrator circuit'
5 List different types of MOSFETs (5*2 =10)

PART B

(Answer any Five. Each question carries 6 marks)

II

1. Draw and explain the fixed bias circuit.
2. Explain the working of a Class B push pull amplifier
3. With a neat sketch, explain the operation of a Hartley oscillator
4. Describe the operation of a bistable multi vibrator
5. Mention the methods of inter stage coupling in multi stage amplifiers
6. Describe the working of a Wein bridge oscillator
7. Describe the constructional details of Depletion type MOSFET (5*6=30)

PART C

(Answer one full question from each module. Each question carries 15 marks)

Module I

III

a) What are the factors to be considered in the selection of operating point? (4)
b) Mention the merits and demerits of fixed bias circuit. (4)
c) Explain with a neat sketch, the operation of Emitter follower and state its Applications (7)

OR

IV

a) With a neat sketch explain the working of a transformer coupled amplifier.

Draw its frequency response (8)

b) What is the need of a multi stage amplifier ? How the gain of multi stage amplifier is calculated (4)

c). A multi stage amplifier consists of three stages. The voltage gains of the stages are 20 dB, 25 dB and 30 dB. Calculate the over all voltage gain ? (3)

Module II

V a) A parallel resonant circuit has a capacitance of 100 pf in one branch and an inductance of 100 micro henry plus a resistance of 10 ohm in parallel branch. If the supply voltage is 10v, calculate resonant frequency, Line current and impedance of the resonant circuit at resonance . (10)

b) State the importance of heat sinks and heat dissipation in power amplifiers (5)

OR

VI a) Explain why a power amplifier is always preceded by a voltage amplifier ? (5)

b) Although a Class B single ended power amplifier has high efficiency, yet it never used in practical circuits. Explain why ? (5)

c) Explain why a step down transformer is used in the output of a power amplifier Circuit ? (5)

Module III

VII a) Draw a crystal oscillator circuit and explain its working principle (6)

b) List the advantages of crystal oscillator (3)

c) Describe the advantages of negative feed back in amplifier (6)

OR

VIII a) Can a negative feed back amplifier work as an oscillator ? if yes, how? If not Why ? (5)

b) Explain why an LC tank circuit once excited, does not produce sustained oscillations ? (5)

c) Classify the different types of negative feed back in amplifiers. (5)

Module IV

IX a) Explain the working of mono stable multi vibrator with a neat sketch and plot its output wave forms. (9)

b) Give the constructional details of N- channel JFET (6)

OR

X Draw the circuit diagram of an astable multi vibrator. Justify that it is a two stage RC coupled amplifier using feed back. How does it generate square waves? (15)

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